

Unstructured Mesh Generation

Technical Progress Report (Weeks 9 & 10)

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FOSSEE Internship – Progress Report

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Project Objectives & Core Design

- ◀ **[Done] cfMesh vs snappyHexMesh:** Better boundary layer generation, fewer parameters, and excellent handling of trailing edges.
- ◀ **[Done] Full Workflow:** Load STL → Create Domain → Mesh → Visualize in Blender.
- ◀ **[Done] Inspection:** Users can zoom into specific areas to inspect mesh quality intuitively.
- ◀ **[Done] Real-time Visualization:** Live visualization feedback during the meshing and simulation process.

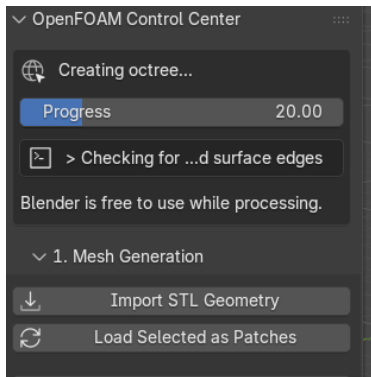
Project 3: Unstructured Mesh with CF-mesh

- **CF-mesh preferred** over SnappyHexMesh (better boundary layer generation, fewer parameters)
- Real-time visualization during meshing
- Load STL → create domain → mesh → visualize in Blender
- Users can zoom into specific areas to inspect mesh quality
- CF-mesh handles trailing edge boundary layers better

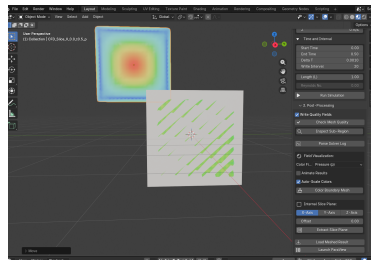
1. Core Features Implemented

- ◀ **Real-Time Visualization:** Integrated continuous visual feedback, enabling users to monitor meshing and solver progression live within the Blender viewport.
- ◀ **Specific Geometry Inspection:** Users can now isolate and view any specific part or region of the generated mesh/geometry to easily debug boundary layers.
- ◀ **Advanced Slice Visualization:** A dedicated *Slice Plane* feature allows internal visualization of flow and mesh cross-sections, providing deep insights into wake and vortex regions.
- ◀ **Dual Solver Support:** Complete integration of both **icoFoam** (transient solver) and **simpleFoam** (steady-state solver) within the UI.

Visualizing New Capabilities



*Real-Time Meshing Update



*Internal Slice Visualization

2. Project Completion & Testing Phase

Milestone Achieved

Almost all planned features are now successfully implemented. The core meshing workflow, OpenFOAM solver integration, and VTK-based Blender visualization are highly functional.

Current Focus: Comprehensive Testing:

- ▶ End-to-end user workflow execution across varied complex STL geometries.
- ▶ Edge-case validation for the automated OpenFOAM dictionary generation.
- ▶ Ensuring UI responsiveness and preventing Blender freezes during heavy I/O operations.

3. Known Issues in Testing

- ▶ **VTK Processing Artifacts:** Occasionally, errors in orthogonality and skewness emerge in specific figures or localized areas of the mesh.
- ▶ **Root Cause Analysis:**
 - Believed to be tied to OpenFOAM's foamToVTK conversion or specific interpolation behaviors near complex boundary intersections.
- ▶ **Ongoing Mitigation:** Actively investigating the VTK parsing logic and considering smoothing algorithms or adjusting OpenFOAM meshing tolerances to eliminate these graphical and topological errors.

week9_vtk_error.png

4. Timeline & Milestones

Week	Task	Status
4	Turbulence models (RANS), Mega exception handling	Done
5 – 6	STL Importer, Jet Colormaps, Residuals & Y+ Calculus	Done
7	System file opener, Custom liquid presets & Smart UI	Done
8	OpenFOAM 2212 migration, Modular Arch, Dynamic Patches	Done
9 – 10	Real-time Vis, VTK Slices, icoFoam & simpleFoam	Done (Testing)
11 – 12	Final Documentation, Bug Fixes & Project Handover	Upcoming

Progress Summary (Weeks 9 & 10)

Feature/Milestone	Outcome/Benefit
Real-Time Visualization Geometry & Slice Views	Live feedback of meshing iterations Inspect interior volume and specific boundary parts
Solver Integration	Native support for icoFoam and simpleFoam
Development Status	Feature-complete , transitioned to test- ing phase
Known Issues	Minor orthogonality/skewness errors dur- ing VTK extraction (Under Investigation)

Thank You!

End of Weeks 9 & 10 Report